

CS216 Everything Data Sp26



Resources

Example Grade Calculator

This is a Google Spreadsheet that, when you click on the link, you will be prompted to make a copy. You can then fill it in as you please. We cannot guarantee that the grade you calculate with this spreadsheet is the grade you will get. This is to help you understand your grade and reduce confusion on how to calculate it.

Technology Install Directions

This class uses Jupyter Notebooks. You can use whatever platform you want to run them. We recommend using [VS Code with the Jupyter extension](#). If you already have VS Code installed, make sure you also have the extension and **update** to at least Python version ≥ 3.7 (preferably newer). When you are done, we recommend opening a Jupyter Notebook, running it, and beginning to familiarize yourself with the [VS Code Jupyter Notebook documentation](#).

Gradescope

We will be using Gradescope to submit assignments. If you are unfamiliar with Gradescope or aren't sure how to submit your assignment, they created a [Gradescope help document for you](#).

Jupyter Notebook Container

If something happens to your computer or you cannot install Anaconda on it, we've reserved containers for you through OIT. Go to the [container manager](#) and look for the **Pytorch**: *JupyterLab with Pytorch for Data Science and Machine Learning* machines. Make sure you select **Pytorch** and not the **Jupyter**: *Python and Julia: data science / scientific computing notebooks* option.

Click on the button to reserve your instance of the notebook. Once your instance is reserved you can click on **Pytorch** among your reserved containers, start the server, and upload any necessary files.

How to properly format/submit your notebooks

Git Mental Model Help

Python for Data Science

If you are new to programming in Python, there are a lot of good tutorials available. The official documentation has one: <https://docs.python.org/3/tutorial/index.html> and google also hosts a good tutorial with videos: <https://developers.google.com/edu/python/>. If you want a guide that is specific to transitioning to Python from Java, try [Python for Java Programmers](#). Note that if you are new to programming altogether, you do not meet the pre-requisites for the class and should consider CS 101 or CS 116 instead; we are assuming that you have the background to pick up basic syntax and functionality of Python on your own.

If you are new to scientific programming with Python, you may find [this NumPy tutorial](#) helpful, along with the [NumPy documentation](#). The [Python Data Science Handbook](#) is also a very useful reference the use of Python for data science, including helpful information on commonly used libraries like NumPy, Pandas, Matplotlib, and Scikit-Learn. For a gentler introduction, try

this online [data8 book developed for U.C. Berkeley's Foundations of Data Science course and used in CS 116 at Duke.](#)

To get all of the data science libraries you need together with a Python distribution on your local device, look at the [Anaconda distribution](#), available for free. The Anaconda distribution of Python contains everything that you need to be successful in data science with Python, including all Python resources you should need for this course. It includes Python 3 itself, all of crucial libraries for data science (NumPy, Pandas, Matplotlib, scikit-learn, etc.), and development environments (notably the Spyder scientific computing IDE and Jupyter notebooks).

Duke Co-Lab

The [innovation Co-Lab](#) hosts a variety of trainings, projects, and programming that might be interesting to an aspiring data scientist. The [Co-Lab also hosts regular office hours \(and you can make an appointment\)](#) on a variety of technical subjects.

Academic Support Resources

The Academic Resource Center (the ARC) offers services to support students academically during their undergraduate careers at Duke. The ARC can provide support with time management, academic skills and strategies, course-specific tutoring, and more. ARC services are available free to any Duke undergraduate student, studying any discipline. (919) 684-5917, theARC@duke.edu, or arc.duke.edu.

Long Term Health Issues, Personal Distress, or Emergencies

If you have or develop a chronic health issue that will interfere with your participation in this course, in a situation of extreme personal distress, or an emergency interferes with your participation in this course, please contact your academic dean to seek accommodations as directed by Trinity College.

Mental Health and Wellbeing Resources

Duke is committed to holistic student wellbeing, which includes one's mental, emotional, and physical health. The university offers resources to help students manage daily stress, to encourage intentional self-care, and to access just-in-time support. If you find you need support, your mental and/or emotional health concerns are impacting your day-to-day activities, your academic performance, or you need someone to talk to, the resources below are available to you:

- [DukeReach](#) – DukeReach provides comprehensive outreach services to support students in managing all aspects of wellbeing, including referrals and follow-up services for students who are experiencing significant challenges related to mental health, physical health, social adjustment, and/or a variety of other stressors. You can contact the DukeReach team at dukereach@duke.edu.
- [Counseling & Psychological Services \(CAPS\)](#) – CAPS offers counseling services to Duke students including virtual appointments, and referrals in the community. You do not need an appointment for an initial assessment. You may walk in or call 919-660-1000 to get started. Hours: Monday-Friday 9:00am – 4:00pm. After hours counseling services are available at no additional cost to students, you can call: 919-660-1000 Option 2
- [TimelyCare](#) (formerly known as Blue Devils Care) – TimelyCare is an online platform that is a convenient, confidential, and free way for Duke students to receive 24/7 mental health support through TalkNow and scheduled counseling.
- [Duke Student Health](#) – Student Health offers a wide range of healthcare services for all Duke students, many of which are covered by the student health fee. To make an appointment call (919) 681-9355. Hours: Monday – Friday, 8am – 4:30pm, Thursday 9am – 4:30pm. Closed from 12-12:30 each day.

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